Charles Bergenske

UC Santa Cruz | Robotics Engineering B.S.

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SUMMARY

Robotics engineering student training legged locomotion policies using asymmetric actor-critic PPO and domain randomization in NVIDIA Isaac Lab. Currently implementing a belief-state policy for blind proprioceptive walking, trained by teacher-student distillation. Second author on an accepted AIAA paper addressing multirotor UAV wind state estimation.

EDUCATION

B.S. Robotics Engineering · University of California, Santa Cruz — Sep 2024 – Jun 2028
GPA 3.91 / 4.00 — *Santa Cruz, CA*

- Coursework: Control Systems, Embedded Systems, Linear Algebra, Data Structures, Kinematics & Dynamics, Probability & Statistics

EXPERIENCE

Student Associate Engineer (Intern) · Rosen Aviation — Jun 2025 – Sep 2025, Jun 2026 – Present
Software Engineering - Test Tools — *Eugene, OR*

- Designed and implemented custom Mask R-CNN instance segmentation model and accompanying physical test harness, fully automating LCD power-cycle testing across 2000 cycles. Annotated ~750 image dataset for training, using COCO annotation format.
- Conducted feasibility study on pre-batch screening process. Saved ~$15,000 and 250 annual labor hours.
- Utilized Altium to design capacitor hold-up PCB layout.
- Designed and built WPF C#/xaml application, automating DC monitor burn-in procedure.

Undergraduate Researcher · Applied Math Department, Baskin Engineering, UC Santa Cruz — Nov 2025 – Present
Advisor: Prof. Javier Gonzalez Rocha — *Santa Cruz, CA*

- Second author on "A Study of the Effect of Vehicle Inertia on sUAS Wind Estimation Performance," accepted for AIAA SciTech 2027.
- Built and instrumented multirotor airframes for flight validation, using battery-bay ballasting to vary mass while preserving airframe geometry.

Undergraduate Researcher · Human Aware Robotics Exploration Lab, UC Santa Cruz — Oct 2025 – Jun 2026
Advisor: Prof. Steve McGuire — *Santa Cruz, CA*

- Trained stationary height controller policy for a Unitree B1 quadruped with skrl PPO and disturbance curriculum, progressively increasing force magnitude, improving robustness to pushes and slips.
- Researched teacher-student distillation techniques in which a student policy learns a belief-state encoder to mimic a teacher policy with access to privileged data, allowing the robot to remain robust in unpredictable circumstances.

PROJECTS

BDX-R Bipedal Locomotion — Jul 2026 – Present
Python, NVIDIA Isaac Lab, C++, ROS2 · github.com/cbergens/BDX-R-IsaacLab-Belief

- Trained blind proprioceptive walking policies in NVIDIA Isaac Lab using asymmetric actor-critic PPO with actuator and rigid-body material randomization across 4096 environments and ~513M timesteps, robust to rough terrain and velocity perturbations up to 1.06 m/s.
- Patched a scene-graph traversal bug in NVIDIA Isaac Lab that applied contact sensors to only 1 of 11 rigid bodies on URDF import, silencing inputs to 3 reward terms. Restored full contact and verified the sim ground reaction force (172N) against the analytical prediction (174N).
- Ported entire training stack from Open Duck Mini v2 model to larger, heavier BDX-R model, implementing Froude similarity to properly scale step gait parameters.
- Extending to teacher-student distilled belief-state policy.

Cascade PID Quadrotor Flight Controller — May 2025 – Jun 2026
MATLAB, CoppeliaSim, control allocation · github.com/cbergens/Drone-Control

- Built a two-loop cascaded flight controller, with outer position control loop driving internal attitude control loop.
- Diagnosed and resolved a diverging spiral yaw-drift bug, traced to a simulation sensor timestep mismatch.

TECHNICAL SKILLS

Languages	Python, C, C#, C++, MATLAB
Robotics	NVIDIA Isaac Sim/Lab, Omniverse, ROS 2, ArduPilot
RL	PPO, rsl_rl, skrl, domain randomization, reward shaping
ML & Vision	PyTorch, TensorFlow, OpenCV
Embedded	STM32, MPLAB
Tools	Linux, Git, CMake, SolidWorks, Altium

AWARDS AND SOCIETIES

Dean's Honors List · Baskin School of Engineering, UC Santa Cruz — 2025 – 2026
2025: Winter, Spring, Fall | 2026: Winter

Tau Beta Pi · University of California, Santa Cruz — Jun 2026 – Present